

1. Introduction

Art. 83 EPC stipulates that the application "shall" (previously "must") disclose the invention in a manner sufficiently clear and complete for it to be carried out by a person skilled in the art. The related ground for opposition is in Art. 100(b) EPC.

The subject-matter of an application must be sufficiently disclosed at the date of the application (see in this chapter II.C.2.), based on the application as a whole (II.C.3.1.), including examples (II.C.5.3.), and taking into account the common general knowledge of the skilled person (II.C.4.). At least one way of enabling the person skilled in the art to carry out the invention must be disclosed (II.C.5.2.), but this is sufficient only if it allows the invention to be performed in the whole range claimed (II.C.5.4.). Parameters must be sufficiently defined (II.C.5.5.). The disclosure must also be reproducible without undue burden (II.C.6.); evidence from post-published documents is allowable under certain circumstances (II.C.6.8.).

The distinction between the requirements for sufficiency of disclosure under Art. 83 EPC and clarity of the claims under Art. 84 EPC (II.C.8) is important, as examination in respect of the requirements of Art. 83 EPC is still permitted during opposition proceedings, whereas in respect of Art. 84 EPC it is limited to cases where there has been an amendment (see also G 3/14 in chapter II.A.1.4.).

The burden of proof generally lies with the opponent to establish that an invention is insufficiently disclosed (II.C.9.).

Sufficiency of disclosure as applied in biotechnology is considered in this chapter II.C.7., including the issue of broad claims (II.C.7.1.4.).

A number of decisions have looked at the monopoly conferred by a patent, how this is linked to the notion of a "technical contribution" and what this means for assessing sufficiency of disclosure; see, for example, the following decisions reported elsewhere in this chapter: T 409/91 (OJ 1994, 653), T 435/91 (OJ 1995, 188), T 1164/11, T 1845/14, T 2015/20, T 797/14 and T 2119/14; compare also with T 416/14 (as regards a proprietor's argument that there were no potential effects on the competition). The issue of where the line is to be drawn between Art. 56 EPC and Art. 83 EPC was recently addressed in e.g. T 409/17 (point 3.3.3 of the Reasons), referring to G 1/03 and T 1845/14; on this point, see also the following decisions: T 380/05, T 1311/15, T 1079/08, T 862/11, T 2001/12, T 898/05, T 967/09, T 116/18 (OJ 2022, A76).

2. Date of compliance

Sufficiency of disclosure under Art. 83 EPC requires inter alia that the subject-matter claimed in a European patent application be clearly identified. This requirement must be complied with as from the date of filing because a deficiency in a European patent application as filed, consisting in an insufficient identification of the subject-matter claimed, cannot subsequently be cured without offending against Art. 123(2) EPC which provides

that the subject-matter content of a European patent application as filed may not be extended (G 2/93, OJ 1995, 275).

It is not the purpose of the patent system to grant a monopoly for technical speculations that cannot be realised at the time of filing (T 1164/11). See also in this chapter II.C.6.8. "Post-published documents". On monopoly, see the discussion in e.g. T 1845/14 and the reference to this principle in T 2015/20.

According to the board in T 512/07, objections under Art. 83 EPC 1973 may under some circumstances be overcome by amending the claims, since thereby the "invention" referred to in Art. 83 EPC may be so changed that it no longer depends on the insufficiently disclosed aspects of the application for its realisation. However such objections cannot be overcome by amendment of the description and drawings, since the amendment would then add subject-matter to the application as filed. In general, the claimed subject-matter at issue had to be examined for compliance with Art. 83 EPC 1973 on the basis of the application documents as originally filed.

3. Parts of the application relevant for assessing sufficiency of disclosure

3.1. Application as a whole

Sufficiency of disclosure within the meaning of Art. 83 EPC must be assessed on the basis of the **application as a whole** – including the description and claims – (see leading decisions T 14/83, OJ 1984, 105; and also T 169/83, OJ 1985, 193) and not of the claims alone (see e.g. T 202/83, T 179/87 of 16 January 1990, T 435/89, T 82/90, T 126/91). The drawings must be considered as ranking equally with the other elements of the application when considering whether the requirements of Art. 83 EPC (and Art. 84 EPC) have been satisfied (see T 169/83, OJ 1985, 193; T 308/90 and T 818/93).

In T 32/84 (OJ 1986, 9) it was pointed out that the fact that certain elements of an invention essential to its operation were not referred to explicitly either in the claims, or in the relevant portion of the description nor shown in the drawing of the invention as claimed did not necessarily mean that the application did not disclose the invention in a manner sufficiently clear and complete for it to be carried out by a person skilled in the art as required by Art. 83 EPC 1973. This decision was cited, inter alia, in T 391/91, T 830/02 and T 25/09.

According to established case law, substantially any embodiment of the invention, as defined in the broadest claim, must be capable of being realised on the basis of the disclosure. This implied in particular that an objection of insufficiency could be raised against the subject-matter of any claim, independent or dependent (R. 29(3) EPC 1973, now R. 43(3) EPC). From a legal point of view it was therefore irrelevant whether or not the feature objected to was essential or to what degree the scope of protection conferred by the patent depended on the claim in question (T 226/85 (OJ 1988, 336), cited in many decisions, for example, T 1011/01 and T 1129/09).

The board in ex parte case T 206/13 could not accept the appellant's submissions to the effect that preferred or optional features defined in a claim should be disregarded in the

assessment under Art. 83 EPC. According to the established case law the requirement of sufficiency of disclosure defined in Art. 83 EPC was only complied with if the disclosure of the invention allowed the skilled person to perform, without undue burden, essentially all the embodiments covered by the claimed invention. This applied in particular to the specific particular embodiments of an invention defined in dependent claims pursuant to R. 43(3) EPC (T 1011/01) and, by the same token, to any optional feature defined in a claim, since such a feature also constituted, by its very nature, a particular embodiment of the claimed invention, irrespective of whether the optional feature was qualified as being "preferred" or not.

In T 797/14 the board came to the conclusion that the essential element of the claimed invention, namely the coating composition Flurotec (TM), was not of public knowledge and that there was also not enough information available to the skilled person for him to reliably determine the composition or structure of the product. The counterpart of a monopoly by a patent is however the disclosure of the invention, in particular of its essential elements, and not the provision or use of a commercial product whose structure and composition are not public. (On disclosure of a **starting material**, see T 1596/16, in which one of the parties had invoked T 2399/10 and T 797/14 but the board, having summarised the issues dealt with in those decisions, found that the case in hand was not comparable.)

Disclosure is insufficient if the skilled person, taking into account the entire teaching of the patent, cannot rework an invention which is defined in the claims in a completely clear and comprehensible manner, unless he disregards a meaningful feature (T 432/10, "Catchword").

3.2. Alleged effect not a feature of the claims

According to the case law of the boards of appeal, the requirement of sufficiency of disclosure relates to the invention defined in the claims, and in particular to the combination of structural and functional features of the claimed invention, and there is no legal basis for extending such a requirement also to encompass other technical aspects possibly associated with the invention (such as results to be achieved or technical effects) mentioned in the description but not required by the claimed subject-matter (principle reiterated in e.g. T 1900/17).

An objection of insufficient disclosure under Art. 83 EPC cannot legitimately be based on an argument that the application would not enable a skilled person to achieve a non-claimed technical effect (T 2001/12, citing mainly G 1/03 (OJ 2004, 413) but also T 1079/08, T 939/92 and T 260/98). Other decisions referring to this principle: T 1744/14, citing T 1345/14, point 9.8 of the Reasons; T 1311/15, point 5.2 of the Reasons; and T 2001/12, point 3.4 of the Reasons; T 1900/17, point 2.5 of the Reasons; T 1943/15; T 1502/16, citing T 862/11.

It is the board's settled case law that a technical effect is to be taken into account in assessing enablement only if it is explicitly claimed. If it is not claimed, it is irrelevant for enablement whether or not it is actually achieved. The question whether it is achieved

might then become relevant when assessing inventive step, namely for the purposes of formulating the objective technical problem (T.1216/12, citing T.862/11 and T.2001/12).

T.1311/15 the parties did not dispute that claim 1 did not include any flammability requirement in its wording. It was however also undisputed that the invention aimed at providing non-flammable refrigerant compositions. The question arose whether, when examining the ground of opposition under Art. 100(b) EPC, the intended effect of the claimed subject-matter should be taken into account (issue dealt with by reference to the aforementioned settled case law).

T.2001/12 dealt with the distinction between the requirements of sufficiency of disclosure (Art. 83 EPC), clarity of the claims (Art. 84 EPC) and inventive step (Art. 56 EPC). That decision was cited in T.862/11, in which the board addressed the relevance of the technical effect for the purposes of Art. 83 EPC and concluded that a distinction had to be made between how the effect was to be considered in that context and how it was to be considered for the purposes of Art. 56 EPC (see T.2210/16 for an analysis of these decisions). On the relationship between an objection under Art. 56 EPC and one under Art. 83 EPC, see also T.1099/16 (use of a known compound performing a new function), in which it was held, in line with G.1/03, that, if an effect is expressed in a claim [as in the case in hand, adhesion enhancer], there is a lack of sufficient disclosure [which was not a ground for opposition in the case in hand]. Otherwise, i.e. if the effect is not expressed in a claim but is part of the problem to be solved, there is a problem of inventive step.

T.206/13 also applied T.2001/12, in so far as the examining division, in the context of the assessment of sufficiency of disclosure of claims 1 and 11, had referred to technical aspects which, in its opinion, were not sufficiently disclosed. The board pointed out that these technical aspects were not defined in claims 1 and 11 and, consequently, were not to be considered in the assessment under Art. 83 EPC of the invention defined in claims 1 and 11. The requirement of sufficiency of disclosure related to the invention defined in the claims, and in particular to the combination of structural and functional features of the claimed invention, and there was no legal basis for extending such a requirement also to encompass other technical aspects possibly associated with the invention (in particular, technical features or effects mentioned in the description) but not required by the claimed subject-matter. Such technical aspects might be pertinent in the assessment of other requirements of the EPC (in particular, the requirements of Art. 84 and 56 EPC).

In T.1744/14, the problem set out in the contested patent and the effect(s) underlying it were not part of claim 1. The board acknowledged that in e.g. T.593/09, the subjective problem to be solved according to the patent, even though not part of the claims, had been taken into account when deciding on sufficiency of disclosure. However, the question in that case was not whether the claimed products objectively actually solved the technical problem, i.e. demonstrated the technical effects mentioned in the patent, but rather whether, in view of the ill-defined parameter present in the claim, products according to the claim could be selected by the skilled person without undue burden, such that a solution to the alleged problem set out in the patent would have been provided. In case T.1744/14, however, the skilled person was able to obtain polymorph II as defined in

claim 1 without any undue burden. Hence, case T 1744/14 was different to that underlying T 593/09.

In T 1469/16 the examining division considered that for the main request to comply with Art. 83 EPC, it was necessary that the skilled person, by carrying out the invention, would have achieved the effect of improved coding performance. However, in the board's view the claim 1 did not express such an effect, and the same applied to the other claims of the main request. This meant that if the effect was not achieved over the whole scope of any of the claims, this was not a problem under Art. 83 EPC but might have an impact on the assessment of inventive step (see T 939/92, OJ 1996, 309; G 1/03, OJ 2004, 413 and T 862/11).

4. Knowledge of skilled person relevant for assessing sufficiency of disclosure

4.1. The disclosure is aimed at the skilled person

The claimed invention must be sufficiently disclosed as from the effective date of filing of the application. This disclosure is aimed at the person skilled in the art who may rely on common general knowledge to supplement the information contained in the patent. Textbooks and general technical literature form part of the common general knowledge, which however does not normally include scientific articles and patent literature.

The same level of skill has to be applied when, for the same invention, the two questions of sufficient disclosure and inventive step are being considered (T 60/89, OJ 1992, 268; T 694/92, T 187/93, T 412/93). However, the standard of disclosure required for amendments by Art. 123(2) EPC 1973, namely that of being directly and unambiguously derivable, is inappropriate, the criterion being that it must be possible to reproduce the invention on the basis of the original application documents without any inventive effort and undue burden (T 629/05; cited in T 79/08).

It is the **same skilled person** that has to be considered when construing the subject-matter of a claim. It accordingly follows that the construction of a particular claim should be identical for the assessment of inventive step and sufficiency of disclosure (T 967/09).

Thus, the skilled person may **use his common general knowledge to supplement the information** contained in the application (T 206/83, OJ 1987, 5; T 32/85, T 51/87, OJ 1991, 177; T 212/88, OJ 1992, 28; T 772/89). He may even recognise and rectify errors in the description on the basis of such knowledge (T 206/83, OJ 1987, 5; T 171/84, OJ 1986, 95; T 226/85, OJ 1988, 336). Textbooks and general technical literature form part of the common general knowledge (T 171/84, T 51/87, T 580/88, T 772/89). Common general knowledge does not normally include patent literature and scientific articles (see T 766/91, point 8.2 of the Reasons; T 1253/04, point 10 of the Reasons; see also T 59/18, in which a single patent was relied on and which refers to T 412/09 (series of patents)). Information which can only be obtained after a comprehensive search is likewise not to be regarded as part of the common general knowledge (T 206/83, T 654/90). According to the board in T 475/88, a disputed claim of common general knowledge must